



# Lecture notes for Applies statistics, Probability theory

## Lecture 10.

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# Presentation Overview

Correlation

Regression  
analysis

Correlation

Covariance

Correlation

Regression analysis

Regression curve

Nonlinear regression



# Section Overview

## Correlation

Covariance

Correlation

Regression  
analysis

Correlation

Covariance

Correlation



## Concept question

The end of the semester is slowly, but surely approaching and You want to know the answer to the following question:

*"Does it make any sense to spend time on preparing for the tests and exams?"*

With other words, how strong effect does the amount of study time have on my results? To answer this question you spent a day collecting data from your senior student friends, and the results were the following:

|                      | S1 | S2 | S3 | S4 | S5 | S6 | S7 | S8 | S9 | S10 |
|----------------------|----|----|----|----|----|----|----|----|----|-----|
| Hours spent studying | 70 | 23 | 4  | 57 | 40 | 34 | 69 | 0  | 18 | 50  |
| Exam result          | 5  | 3  | 2  | 4  | 4  | 3  | 5  | 1  | 1  | 4   |



Covariance describes the directional relationship between two random variables.

## Definition

The covariance of random variables  $X$  and  $Y$  is

$$\text{cov}(X, Y) = \mathbb{E}((X - \mathbb{E}X)(Y - \mathbb{E}Y))$$

## Remark

The covariance is not a normalized quantity, by normalizing the covariance we get the correlation.



# Correlation

Correlation

Covariance

Correlation

Regression

analysis

Correlation is the statistical relation between two or more variables such that systematic changes in the value of one variable are accompanied by systematic changes in the other.

## Definition

The correlation coefficient of random variables  $X$  and  $Y$  is

$$\text{corr}(X, Y) = \frac{\mathbb{E}((X - \mathbb{E}X)(Y - \mathbb{E}Y))}{\mathbb{D}X \cdot \mathbb{D}Y} = \frac{\mathbb{E}(X \cdot Y) - \mathbb{E}X \cdot \mathbb{E}Y}{\mathbb{D}X \cdot \mathbb{D}Y}$$



# Correlation

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## Remark

If  $X$  and  $Y$  are independent, then  $\text{corr}(X, Y) = 0$ , but from the other side  $\text{corr}(X, Y) = 0$  does not imply, that  $X$  and  $Y$  are independent.

If  $\text{corr}(X, Y) = 0$ , we say, that  $X$  and  $Y$  are uncorrelated.

$-1 \leq \text{corr}(X, Y) \leq 1$ , if  $|\text{corr}(X, Y)|$  is close to one, then the connection considered to be strong.



## Definition

The following quantity is called the empirical correlation coefficient of the  $X$  and  $Y$  samples

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\left[ \sum_{i=1}^n (X_i - \bar{X})^2 \right] \cdot \left[ \sum_{i=1}^n (Y_i - \bar{Y})^2 \right]}}$$

This formula is used to calculate the





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```
x=[70 23 4 57 40 34 69 0 18 50];
```

```
y=[5 3 2 4 4 3 5 1 1 4];
```

```
corrcoef(x,y)
```

```
ans =
```

```
1.0000 0.9324
```

```
0.9324 1.0000
```



# Section Overview

Correlation

**Regression  
analysis**

Regression curve

Nonlinear regression

Regression analysis  
Regression curve  
Nonlinear regression



# Concept question

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## Concept question

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# Concept question - Scatter plot

Correlation

Regression  
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Nonlinear regression

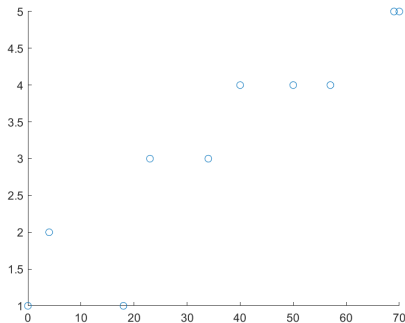


Fig. 1: Scatter plot

```
x=[70 23 4 57 40 34 69 0 18 50];  
y=[5 3 2 4 4 3 5 1 1 4];  
scatter(x,y)
```



# Regression analysis

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When performing a regression analysis, we are modeling the connection between two or more random variables.

Based on the properties of the regression model, we distinguish two types of regressions: linear and nonlinear.

## Definition

Let  $y = f(x_1, x_2, \dots, x_{p-1})$  be a function with  $p - 1$  variables. The  $x_1, x_2, \dots, x_{p-1}$  variables are called predictors and  $y$  is called the outcome or response variable.

## Remark

If, in a given problem, we use  $p$  amount of variables and all of them are random variables, any of them can be used as the response variable, and the rest  $p - 1$  variables can be used as predictors.



# Regression analysis

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With the help of the methods of regression analysis we can solve 3 types of problems:

## First type

Problems, where the function describing the connection is known and the task is to find the still unknown parameters.

## Second type

Problems, where we investigate the dependence of a random variable from one or more predictors and the goal is to approximate the function that describes the connection.

## Third type

Problems, where we investigate the stochastic connection between two or more random variables.



# Regression curve

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When we got an  $(X, Y)$  random variable pair, we might want to know how can be  $Y$  approximated by an  $f(X)$  function, also how can we find the best approximation in the sense of the least square method. Our task is to find the minimum of:  $\mathbb{E}(Y - f(X))^2$ .



# Regression line

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Let  $X$  and  $Y$  be not independent random variables. We will approximate the value of  $Y$  with the help of a linear function of  $X$ :

$$Y \approx aX + b$$





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$$Y \approx aX + b$$

Our task is to find the constants  $a$  and  $b$ . When we perform an approximation the error can be calculated from the real value of  $Y$  and the approximation:

$$Y - (aX + b)$$



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$$Y - (aX + b)$$

To find the value of  $a$  and  $b$  we have to find the minimum of the following expected value:

$$E(Y - aX - b)^2$$



# Concept question

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## Concept question

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```
x=[70 23 4 57 40 34 69 0 18 50]';
```

```
y=[5 3 2 4 4 3 5 1 1 4]';
```

```
plot(fitlm(x,y))
```



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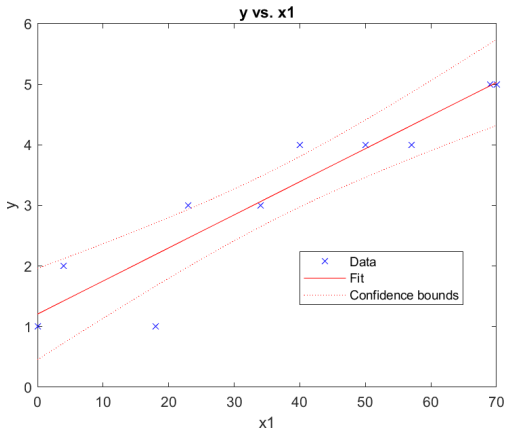


Fig. 2: Line fitting



# Nonlinear regression

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Nonlinear regression is a type of regression analysis, where observational data are modeled by a function which is a nonlinear combination of the model parameters and depends on one or more independent variables.

A nonlinear regression model can be expressed as:

$$Y = f(x, \beta) + \epsilon$$

Where:

- $f(x, \beta)$  is the regression function
- $X$  is the vector of independent variables
- $\beta$  is the vector of parameters that the model aims to estimate
- $\epsilon$  is the error



# Nonlinear regression

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Some frequently used nonlinear regression Algorithms:

- **Polynomial Regression** - a type of nonlinear regression that fits a polynomial function to the data.
- **Exponential Regression** - a type of nonlinear regression that fits an exponential function to the data.
- **Logarithmic Regression** - a type of nonlinear regression that fits a logarithmic function to the data.
- **Power Regression** - a type of nonlinear regression that fits a power function to the data.
- **Gauss-Newton Algorithm** - an iterative optimization method designed for minimizing the sum of squared differences between observed and predicted values in nonlinear least squares regression.
- **Gradient Descent algorithm** - a widely used iterative optimization technique for finding the minimum of a function.



# Polynomial Regression

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The general form of a polynomial regression model is:

$$y = \beta_0 + \beta_1 X^1 + \beta_2 X^2 + \dots + \beta_n X^n$$

Where:

- $y$  - dependent variable
- $X$  - independent variable
- $\beta_0, \beta_1, \dots, \beta_n$  - parameters of the model
- $n$  - degree of the polynomial



# Application

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In 1981,  $n = 78$  bluegills were randomly sampled from Lake Mary in Minnesota. The researchers (Cook and Weisberg, 1999) measured and recorded the data visualized below:

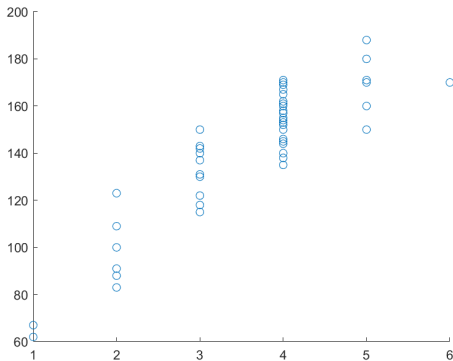


Fig. 3: Bluegill, age in years and length in mm-s





# Application

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Matlab solution:

```
p=polyfit(age,lengthmm,2);  
hold on  
plot(age,lengthmm,'o')  
plot(linspace(1,6),  
polyval(p,linspace(1,6)))
```

